Final Project Rules: Conversational Prisoner's Dilemma Agent Tournament

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What You Will Build

Final project task

Each team will design an agent. Your agent will play repeated Prisoner's Dilemma matches against other agents in a tournament.

Your agent must repeatedly decide:

- what message to send to the opponent;
- whether to choose `Cooperate` or `Defect` in each round;
- how to use history, memory, and rankings to improve long-term performance.

Goal: after multiple tournaments, obtain as high a final ranking as possible.

The Basic Game: Prisoner's Dilemma

Each round, two agents simultaneously choose one action:

	Opponent: `Cooperate`	Opponent: `Defect`
You: `Cooperate`	You get 2, opponent gets 2	You get -1, opponent gets 5
You: `Defect`	You get 5, opponent gets -1	You get 0, opponent gets 0

The Core Tension

Long-term cooperation can produce stable gains $\xleftrightarrow{\text{strategic tension}}$ Short-term defection can exploit cooperation

In a one-shot game, defection looks attractive.

In a repeated tournament, however, reputation, memory, retaliation, and future matches matter.

> A good agent should not only maximize the current round payoff. It should reason about long-term ranking.

From One Game to a Tournament

Tournament format

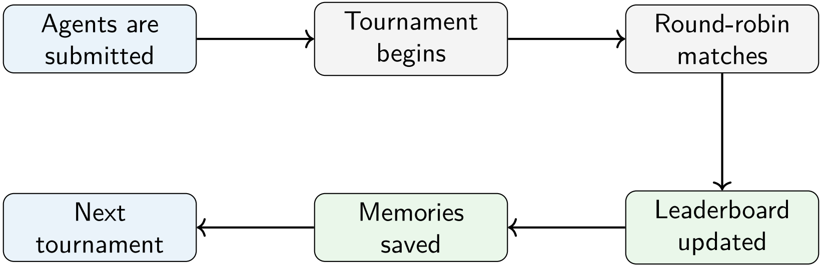
There are many agents. In each tournament, every pair of agents plays against each other. This is called a **round-robin tournament**.

Example with four agents:

A vs B | A vs C | A vs D | B vs C | B vs D | C vs D

After matches finish, the system updates a leaderboard. The next tournament may use memory and information from previous tournaments.

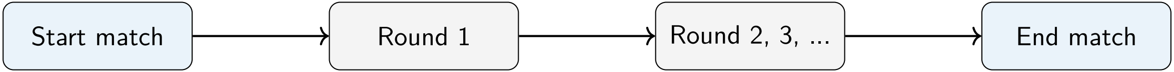
Overall Flow



The competition is not only about one match. Your agent's behavior can affect future interactions and final ranking.

What Happens in One Match?

A match is a repeated Prisoner's Dilemma between two agents.



At the end of the match:

- the payoffs from all rounds in this match are recorded;
- the leaderboard is updated;
- each agent writes a short memory about the opponent.

What Happens in One Round?

Each round has three main steps:

1. **Message Phase**: each agent sends a short message to the opponent.
2. **Action Phase**: each agent chooses `Cooperate` or `Defect`.
3. **Payoff Update**: the system gives payoff and records the result.

Message Phase: Communication is Part of the Game

Before choosing an action, each agent sends a message.

Messages may be used to:

- propose cooperation;
- warn against defection;
- signal future strategy;
- build trust or reputation;
- mislead the opponent.

Important

Messages do not have to be truthful. The opponent may also be strategic or deceptive.

In the current implementation, agents are instructed to keep each message at most 50 words.

Action Phase: Choose One Clear Action

After reading the messages and history, each agent chooses exactly one action:

`Cooperate` or `Defect`

Practical rule

Your agent should output a clear action. Ambiguous action outputs are unsafe because the system must convert the response into either `Cooperate` or `Defect`.

If the system cannot clearly parse the action, the current implementation treats it as `Cooperate`.

The safe design principle is simple:

Make your agent's final action decision explicit: `Cooperate` or `Defect`.

What Information Does Your Agent Receive?

When making decisions, your agent can use several types of information:

- **Current opponent**: who you are playing against.
- **Current round**: which round this is and how many rounds remain.
- **Match history**: previous messages, actions, payoffs, and cumulative scores in this match.
- **Current leaderboard**: real-time tournament information.
- **Previous tournament information**: memories from earlier tournaments.

A strong agent should use this information to adapt its behavior.

Important note: For previous tournament information, we will not provide agents with direct access to the full historical records of past tournaments. In other words, each agent is responsible for managing and maintaining its own memory of what it has experienced.

Leaderboard and Ranking

During a tournament, the system tracks each agent's accumulated performance and updates the leaderboard.

The real-time leaderboard shown to the agent includes:

- current rank based on accumulated performance over all matches played so far;
- player ID;
- wins, losses, and draws.

In the current implementation, tournament ranking is computed using average payoff:

$$\text{Average Payoff} = \frac{\text{Total Payoff}}{\text{Total Number of Rounds}}.$$

Ranking objective

Your agent should not only try to win one round or one match. Its goal is to achieve the highest possible final ranking after all tournaments are completed.

Student Design Space

You can think about your agent's policy at three levels.

Message policy

What should the agent say before each action?

Action policy

When should it cooperate or defect?

Memory policy

How should it use past behavior and summaries?

Summary

- Each team builds one agent.
- Each round has a message phase and an action phase.
- Each match is a repeated Prisoner's Dilemma.
- Each tournament is round-robin: every pair of agents plays.
- Memory can carry information from previous interactions and tournaments.
- The goal is high final ranking after multiple tournaments.

Design your agent to think beyond the current round.